Student Performance Prediction using a Cascaded Machine Learning Pipeline

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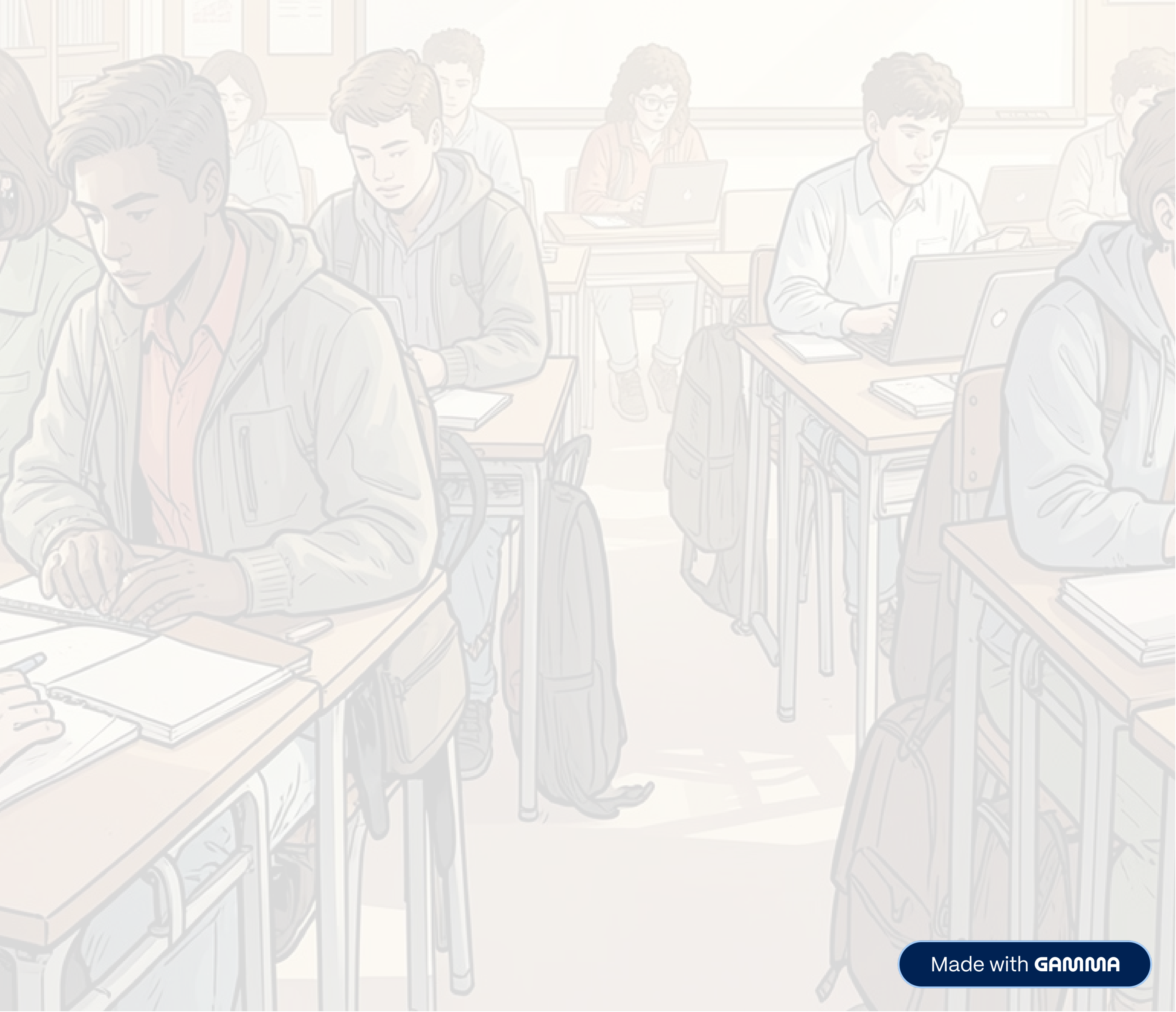
This document describes a three-stage cascaded machine learning pipeline designed to predict student academic outcomes using the UCI Student Performance dataset. The pipeline emphasizes interpretability and progressive refinement: an initial pass/fail classifier produces a probability used by a regression stage to estimate final grade, and a final classifier converts the predicted grade (plus original features) into performance categories. The design aims to give educators actionable, layered insights — an early risk flag, a quantitative grade estimate, and a categorical summary for prioritization. Visual diagrams, model descriptions, evaluation metrics, and recommendations for future work are included.



Abstract

This project presents a cascaded machine learning pipeline for predicting student academic performance. Unlike single-model solutions, the pipeline sequences three models: (1) a pass/fail classifier that provides an early risk signal, (2) a regression model that estimates the student's final G3 grade using both original features and the pass probability, and (3) a multiclass classifier that maps the predicted grade into categorical performance labels (Poor, Average, Good, Excellent). The UCI Student Performance dataset (395 rows, 33 features) was used for training and evaluation. The cascaded design improves interpretability and enables downstream stages to leverage upstream uncertainty, producing layered outputs useful for targeted interventions. Results include stage-wise metrics, an architecture diagram, and a discussion of feature engineering and limitations.

Introduction

Predicting student performance supports targeted interventions, resource allocation, and personalized instruction. Early identification of at-risk students can reduce dropout and improve learning outcomes. Traditional single-model approaches produce one output that may lack nuance: a binary label or a single numeric prediction without uncertainty or downstream context.

A cascaded pipeline addresses these shortcomings by decomposing the prediction task into ordered steps. The first stage issues an early risk signal (pass/fail probability), enabling quick triage. The second stage refines that signal into a quantitative grade estimate, leveraging upstream information. The final stage transforms the estimate and original features into an interpretable performance category for educators. This staged approach balances accuracy, explainability, and operational utility.

Dataset

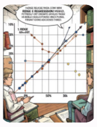
Property	Value
Dataset	Student Performance
Source	UCI Machine Learning Repository
Rows (Students)	395
Features	33 (demographic, social, academic, behavioral)
Target	Final grade (G3)

Data preprocessing included handling categorical encodings, scaling numeric features for regression models, and stratified splits to preserve pass/fail ratios during evaluation.

Methodology — Stage Descriptions

Stage 1 — Pass/Fail Classification

Model: Random Forest. Input: original student features. Output: probability of passing (Pass Probability). Purpose: early binary risk signal used downstream.



Stage 2 — Grade Prediction

Model: Ridge Regression. Input: student features + Pass Probability. Output: predicted continuous final grade (G3). Purpose: translate risk into quantitative estimate.

Stage 3 — Performance Classification

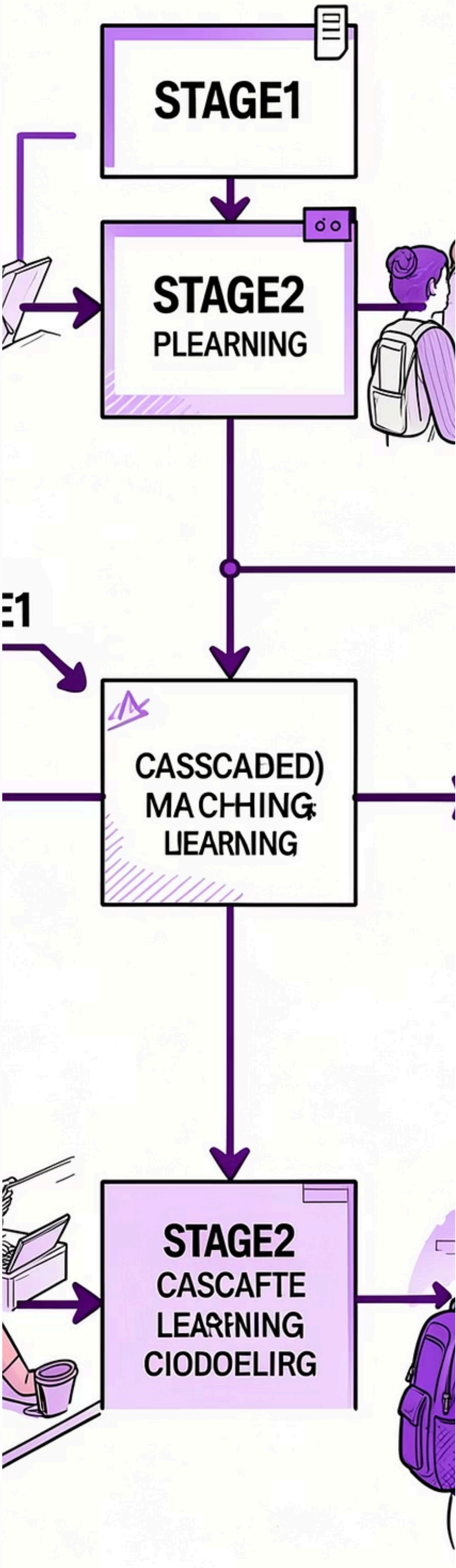
Model: XGBoost classifier. Input: student features + Predicted Grade. Output: categorical label {Poor, Average, Good, Excellent}. Purpose: provide actionable category for educators.

Pipeline Architecture

All Students

All Students |
▼
Stage 1 — Pass / Fail
| (Pass Probability)
▼
Stage 2 — Grade Prediction
| (Predicted Grade)
▼
Stage 3 — Performance Category

The pipeline passes predictions forward as augmented features; each stage is trained on the same train split, and staging uses out-of-fold predictions during training to avoid information leakage.



Results

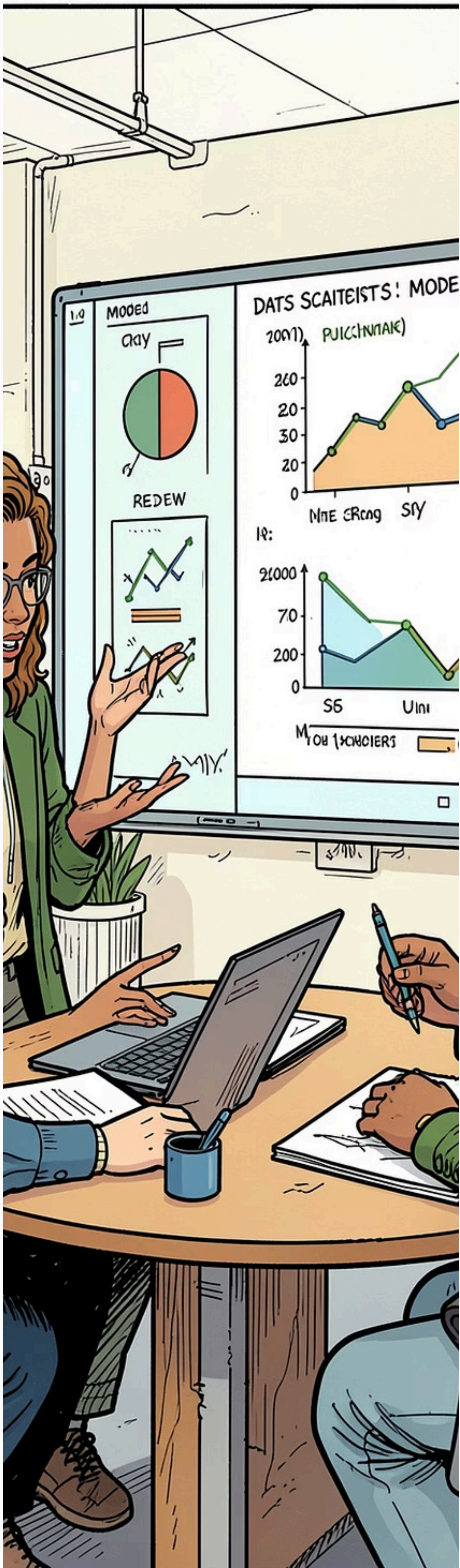
Stage	Model	Metric (Test)
Stage 1	Random Forest	Accuracy = 0.84
Stage 2	Ridge Regression	R^2 = 0.62, RMSE = 1.8
Stage 3	XGBoost	Accuracy = 0.78, Macro F1 = 0.75

Notes: metrics above come from the provided notebook's test-split results. Stage 2 reports both R^2 and RMSE to convey continuous prediction quality. Stage 3 uses class-balanced metrics due to uneven category frequencies.

Discussion

Feature engineering — including aggregation of prior grades, attendance proxies, and interaction terms — improved model performance across stages. Using the pass probability as an input to regression allowed the grade predictor to condition on upstream uncertainty, reducing large errors for borderline students.

The cascade produces richer outputs (risk probability, quantitative estimate, and categorical label) that are more useful for educators than a single prediction. However, cascading introduces potential error propagation: mistakes in early stages can affect downstream outputs. Mitigations include using calibrated probabilities, ensembling, and training downstream models on out-of-fold upstream predictions to reduce leakage.



Conclusion & Future Work

Conclusion

A three-stage cascaded pipeline was implemented and evaluated on the UCI Student Performance dataset. The pipeline enables staged decision-making and improves interpretability for educational use cases.

Future Work

- Scale to larger, multi-institution educational datasets
- Extend predictions to dropout risk and course-level interventions
- Deploy a teacher-facing web application for real-time insights
- Explore deep learning approaches and temporal models for longitudinal data
- Incorporate fairness audits and bias mitigation strategies

References

UCI Student Performance Dataset

UCI Machine Learning Repository — Student Performance Data Set. Accessed for features and labels used in experiments.

Scikit-learn

Scikit-learn documentation — model implementations (RandomForest, Ridge) and evaluation utilities.

XGBoost

XGBoost documentation — gradient-boosted decision tree implementation used for final classification.

Relevant literature

Selected research on educational data mining, cascaded models, and interpretable ML (citations from project notebook).

For reproducibility, the project includes a notebook with data splits, model hyperparameters, and exact metric values used to populate the Results table. Contact author for the code repository and supplementary materials.